

AML — Favorable-Risk Cytogenetics & Mutations



1. FAVORABLE banner

WHAT YOU SEE

Gold 'FAVORABLE' banner arching over the forge.

WHAT IT ENCODES

This whole scene is the favorable-risk ELN 2022 group: core-binding factor (CBF) AMLs plus NPM1-mutated and biallelic CEBPA. These patients have the best complete-remission and overall-survival rates and generally do NOT need allogeneic transplant in first remission.

BOARD TRAP

Favorable risk does not mean 'no consolidation.' They still get cytarabine-based consolidation — they just avoid upfront allo-HCT, which is reserved for intermediate/adverse disease.

2. CBF forge t(8;21)

WHAT YOU SEE

'LEFT STATION — CBF FORGE' with an anvil and '8;21'.

WHAT IT ENCODES

Core-binding factor AML. t(8;21) creates RUNX1-RUNX1T1 fusion. Treated with intensive cytarabine consolidation; high-dose ara-C especially benefits CBF disease. Favorable risk.

BOARD TRAP

t(8;21) is favorable, but do not confuse with t(9;22) BCR-ABL (CML/Ph+ ALL) or with t(15;17) APL — different blots entirely.

3. Auer rod sword

WHAT YOU SEE

A sword labeled 'AUER ROD SWORD'.

WHAT IT ENCODES

Auer rods are crystallized azurophilic granules pathognomonic for myeloid lineage (AML), often abundant in CBF and APL. They confirm the blast is myeloid, not lymphoid.

BOARD TRAP

Auer rods = myeloid, never seen in ALL. If a stem says lymphoblast with Auer rods, the lineage call is wrong.

4. inv(16) eosinophils

WHAT YOU SEE

Purple grapes labeled 'EOSINOPHILS' and 'inv' near 'EOSINOPHILS 33'.

WHAT IT ENCODES

inv(16)/t(16;16) gives CBFB-MYH11 fusion → AML with abnormal marrow eosinophils (AML M4Eo). Favorable risk, same CBF family as t(8;21). KIT mutation can worsen prognosis within CBF.

BOARD TRAP

Abnormal eosinophils point to inv(16), not to chronic eosinophilic leukemia or hypereosinophilic syndrome — context is acute blasts.

5. 55% CR

WHAT YOU SEE

Big '55%' plaque at the CBF station.

WHAT IT ENCODES

Reflects the high complete-remission / favorable outcome rate for CBF AML with intensive chemotherapy. CBF AMLs are the most chemo-sensitive subset.

BOARD TRAP

This is a remission/outcome figure, not a blast percentage. The 20% blast threshold defines AML diagnosis — different number.

6. NPM1 crown forge

WHAT YOU SEE

'CENTER STATION — NPM1 CROWN FORGE' with a king and crown.

WHAT IT ENCODES

NPM1 mutation (nucleophosmin) is favorable WHEN FLT3-ITD is absent or low. Normal karyotype AML; the 'crown' = nucleophosmin mislocalized to cytoplasm. Strong MRD marker by PCR.

BOARD TRAP

NPM1 is favorable only without high-allelic-ratio FLT3-ITD. NPM1-mut + FLT3-ITD high shifts toward intermediate/adverse — the FLT3 red chain cancels the crown.

7. FLT3 red chain

WHAT YOU SEE

A red chain labeled 'FLT3' with 'BLOCKED / FAVORABLE' sign.

WHAT IT ENCODES

FLT3-ITD is the modifier. Low allelic ratio with mutated NPM1 stays favorable; high-ratio FLT3-ITD is adverse. Add midostaurin (FLT3 inhibitor) to 7+3 for FLT3-mutated AML.

BOARD TRAP

Don't reflexively call any FLT3 'adverse.' It is the NPM1 status plus FLT3-ITD allelic ratio together that set the risk; FLT3-TKD differs from FLT3-ITD.

8. MRD ticker tape

WHAT YOU SEE

'PCR TICKER TAPE' with 'MRD' tags and '2 CYCLES'.

WHAT IT ENCODES

NPM1 and CBF fusions are tracked as measurable residual disease by RT-qPCR. MRD positivity after consolidation predicts relapse and can re-trigger consideration of allo-HCT even in favorable patients.

BOARD TRAP

MRD by PCR (molecular) is more sensitive than flow cytometry for NPM1/CBF; a flow-negative but PCR-positive patient is still MRD-positive.

9. 60% / CEBPA

WHAT YOU SEE

'60%' plaque near the right station.

WHAT IT ENCODES

Outcome figure for the CEBPA-mutated favorable group, comparable to NPM1. Reinforces these molecular subsets do well with chemotherapy alone in CR1.

BOARD TRAP

This 60% is prognostic, not a flow MRD cutoff — don't conflate outcome percentages with diagnostic thresholds.

10. CEBPA zipper forge

WHAT YOU SEE

'RIGHT STATION — CEBPA ZIPPER FORGE' with a green zipper.

WHAT IT ENCODES

Biallelic (or in-frame bZIP) CEBPA mutation is favorable risk. The 'zipper' = basic leucine-zipper (bZIP) domain. Normal-karyotype AML with good outcomes on standard therapy.

BOARD TRAP

Only biallelic / bZIP in-frame CEBPA is favorable; a single monoallelic CEBPA mutation does not carry the same favorable prognosis.

11. bZIP label

WHAT YOU SEE

Green zipper segment labeled 'bZIP'.

WHAT IT ENCODES

ELN 2022 refined the CEBPA criterion to in-frame mutations in the bZIP region as the favorable marker, replacing the older 'double-mutant' definition.

BOARD TRAP

Board updates: the favorable CEBPA criterion is now bZIP in-frame, not simply 'CEBPA double mutant' as in older sources.

12. NO HCT sign

WHAT YOU SEE

Wooden sign 'NO HCT' with a crossed bone.

WHAT IT ENCODES

Core teaching: favorable-risk AML in CR1 is consolidated with chemotherapy and does NOT proceed to allogeneic transplant upfront, because transplant toxicity outweighs benefit when relapse risk is low.

BOARD TRAP

Allo-HCT in CR1 is for intermediate (case-by-case) and adverse risk. Sending a favorable t(8;21)/inv(16)/NPM1-only patient to transplant in CR1 is the wrong answer.

13. Myeloid sarcoma

WHAT YOU SEE

Green glowing 'MYELOID SARCOMA' mass with 'SPINE' and 'CRACK'.

WHAT IT ENCODES

Extramedullary myeloid sarcoma (chloroma) can accompany CBF AML, especially t(8;21). It is treated as systemic AML, not as an isolated solid tumor.

BOARD TRAP

Isolated myeloid sarcoma still counts as AML and warrants systemic induction — local radiation alone is insufficient.
